Hao Yu

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Research profile

I study how precursor chemistry and confined vapour transport shape low-dimensional materials. My work connects close-space growth experiments with independent SEM/EDS characterisation, Raman and photoluminescence spectroscopy, and Python analysis.

Education

University of Cambridge · PhD in Engineering | Oct 2022 - present

Thesis submitted 30 September 2026.

Close-space sublimation as a platform for accelerated process discovery in low-dimensional material growth.

Hofmann Group, Department of Engineering. Supervisor: Professor Stephan Hofmann.

University of Oxford · MEng in Materials Science | Oct 2018 - Jul 2022

Trinity College. Research project on photonic micro-ring resonator tuning.

Selected research contributions

Close-space growth of tungsten oxide and sulphide nanostructures | Feb 2023 - Jan 2025

I designed and conducted CSS-facilitated CVD growth and Raman/PL measurements, and co-wrote the first-author paper with Stephan Hofmann. My thesis also records my ex-situ SEM measurements and analysis of collaborator-acquired operando SEM images. Operando SEM, TEM/EDX and XPS acquisition were performed by collaborators. Together, the growth and characterisation evidence links precursor delivery and growth conditions with material formation.

NaCl-assisted close-space growth of WS2 | Jun 2025 - Apr 2026

I investigated salt-assisted close-space growth through paired source-substrate analysis and Raman/PL measurements. I independently acquired SEM-EDS spectra and elemental maps on NaCl-treated SiO2/Si controls to verify salt deposition before growth. This work forms part of my submitted thesis; a first-author manuscript is in preparation.

Optical characterisation of directly written MoS2 | Jun - Aug 2025

I contributed Raman/PL measurements and mapping to a collaborative closed-loop MoS2 writing study, assessing local variations in layer structure and optical response. Other co-authors led the growth platform and complementary measurements. The co-authored manuscript is under review.

RamanPL_2D | Research software

I developed a Python toolkit for Raman and photoluminescence preprocessing, peak fitting, spatial mapping, batch workflows and metadata-aware export.

Photonic resonance trimming | Sep 2021 - Jun 2022

My MEng research concerned micro-ring resonator tuning. I contributed measurements, design and simulations to the subsequent Nano Letters paper on photonic-resonance trimming.

Research outputs

Published journal articles

Hao Yu et al. (first author). A close-space sublimation approach to tungsten oxide and sulfide nanostructure formation. Nanoscale 17(36), 21070-21082. (2025).

First author. Designed and conducted CSS-CVD growth and Raman/PL measurements; co-wrote the manuscript with Stephan Hofmann.

10.1039/D5NR01458A

Sunvir Sahota et al., including Hao Yu. Se nanowire crystal formation via oxidation of 2D HfSe2: A solid-state, in situ reaction coupling for heterogeneous integration technologies. ACS Applied Nano Materials 8(15), 7608-7615. (2025).

Co-author. Raman/SEM acquisition and analysis; SEM imaging on charging-prone substrates.

10.1021/acsanm.5c00308

Nikolaos Farmakidis, Hao Yu et al. (second author). Scalable high-precision trimming of photonic resonances by polymer exposure to energetic beams. Nano Letters 23(11), 4800-4806. (2023).

Second author. Measurements, design and simulations. Associated correction: Nano Letters 24(41), 13103-13104 (2024).

10.1021/acs.nanolett.3c00220

Associated correction: 10.1021/acs.nanolett.4c03363

Minfei Jian et al., including Hao Yu. How do microplastics adsorb metals? A preliminary study under simulated wetland conditions. Chemosphere 309, Part 1, 136547. (2022).

Co-author. Published CRediT contribution: data analysis.

10.1016/j.chemosphere.2022.136547

Under review: Closed-loop crystal layer writing of atomically thin semiconductor

Co-authored manuscript. Contribution: Raman/PL measurements and analysis.

In preparation: Salt-assisted close-space growth of monolayer WS2

First-author manuscript in preparation on source-transport-growth relationships. Topic description; final title to follow.

Research software: RamanPL_2D

Python toolkit for Raman/PL preprocessing, peak fitting, spatial mapping, batch workflows and metadata-aware export. BSD 3-Clause License.

Experience

Investment Analyst Intern · InnoAngel Fund | Jun - Aug 2026

Completed the Little Angel Programme, with experience in early-stage deep-tech investment, sector research and technology assessment. Conducted compute-in-memory chip sector research covering technical routes, value chains and application scenarios.

EnterpriseTECH Analyst · Cambridge Judge Business School | Jan - Apr 2026

Conducted market sizing, application analysis and competitor mapping for photonic photovoltaic technology targeting indoor solar and ambient IoT. Supported a TRL 2-to-TRL 4 roadmap and IP licensing strategy, including MVP and scale-up considerations.

School Ambassador · BOE Technology Group | Jan 2023 - Dec 2025

Organised online and in-person career events at UK universities and identified potential research collaborators for MicroLED technology.

Selected honours and funding

Cambridge Philosophical Society Studentship | 2025-2026

Exhibition Scholar · Trinity College, Oxford | 2020-2022

Singapore Ministry of Education SM1 scholarship | 2014-2017